Unit 5 Questions

1. Explain the **block diagram of regulated power supply (RPS)**.

An electronic circuit that produces a stable DC voltage of fixed value across the load terminals irrespective of changes in the load is known as **regulated power supply (RPS)**. Thus, the primary function of a regulated power supply is to convert an AC power into a steady DC power. The regulated power supply is sometimes also called as a **linear power supply.**

The block diagram of a typical regulated power supply is shown in Figure.

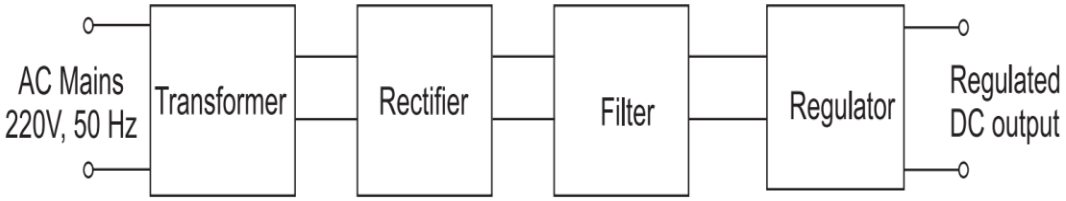


Figure (6.18): *Regulated Power Supply*

The regulated power supply consists various parts, such as

(i)transformer, (ii)rectifier, (iii) filter, and (iv) regulator

(i) **Transformer:** A transformer modifies the alternating current mains voltage to a required value, and its primary duty is to step up or step down the voltage. A step-down transformer, for example, is used in a transistor radio, while a step-up transformer is used in a CRT. Transformers provide separation from the power line and must be used even when voltage changes are not necessary.

(ii) Rectifier: The rectifier circuit is used to convert the input AC voltage into a DC voltage. **The input to a rectifier is AC whereas its output is pulsating DC**. Rectification process can be carried out for the entire cycle or only for half-cycle. Broadly there are two types of rectifiers such as **(i)half-wave rectifier, and (ii)fullwave rectifier**. Full-wave rectifier is used for full wave rectification and half-wave rectifier provides rectification for half-cycle.

iii) Filter: The rectifier's output is pulsating direct voltage with large ripple content and hence, the rectifier's output is undesirable. **A filter is used to provide a pure ripple-free direct voltage**. The filter circuit converts the pulsating direct voltage into a constant direct voltage with almost zero ripple content. Different types of filter circuits such as **capacitor (C) filter, choke input (L) filter, CLC filter and LC filter**.

(iv) Regulator: The output voltage may change or fluctuate due to any change in the **input ac voltage** or the **change in load** or **change in physical parameters** such as temperature of the circuit. This problem can be eliminated with the help of a regulator. A regulator maintains the output constant even when changes at the input voltage or any other changes occur. Different types of regulator circuits such as **transistor in series, Zener diode operating in Zener region and IC regulators** are available.

2. Explain the operation of **full wave rectifier** & draw **wave forms & derive the expression for average output voltage and current**.

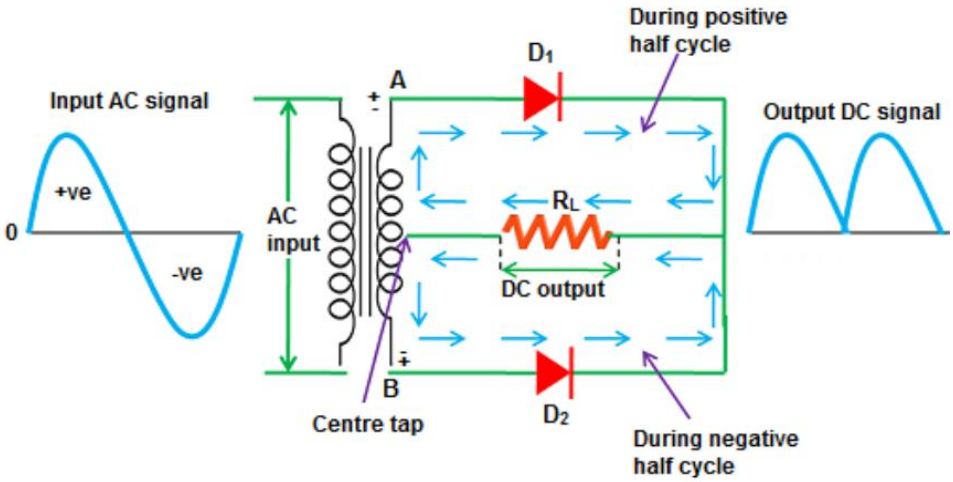
The process of converting the AC current into DC current is called rectification. Rectification can be achieved by using a single diode or group of diodes. A full wave rectifier is a type of rectifier which converts both half cycles of the AC signal into pulsating DC signal. It uses two diodes of which one conducts during one half-cycle while the other diode conducts during the other half-cycle of the applied a.c. voltage.

There are two types of full-wave rectifiers viz.

(i) Full-wave rectifier with center tapped transformer
(ii) Full-wave rectifier without transformer (Bridge rectifier).

i) Centre-Tap Full-Wave Rectifier

A center tapped full wave rectifier is a type of rectifier which uses a center tapped transformer and two diodes to convert the complete AC signal into DC signal. The circuit employs two diodes D1 and D2. A centre tapped secondary winding AB is used with two diodes connected so that each uses one half-cycle of input a.c. voltage. In other words, diode D1 utilises the a.c. voltage appearing across the upper half (OA) of secondary winding for rectification while diode D2 uses the lower half winding OB. The center tap is generally considered as the ground point or the zero voltage reference point.



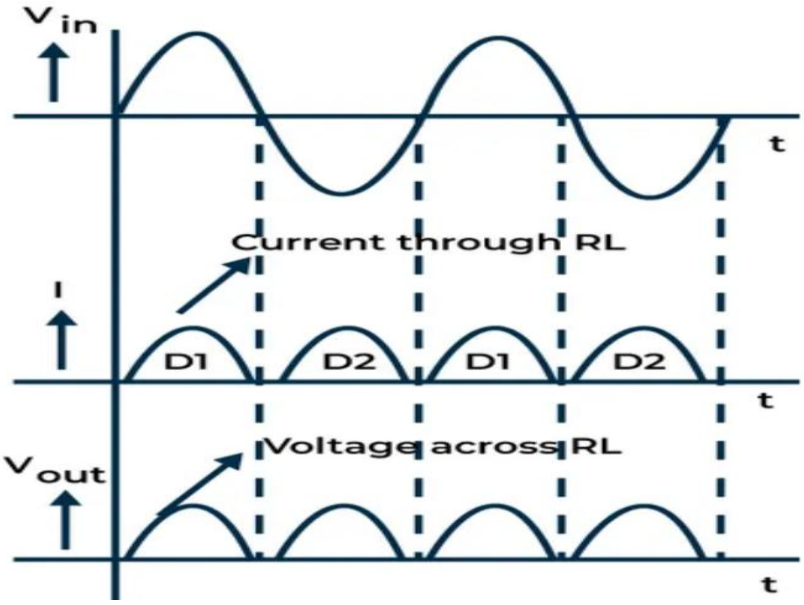
Working operation

- When input AC voltage is applied, the secondary winding of the center tapped transformer divides this input AC voltage into two parts: positive and negative.
- During the **positive half cycle** of the input AC signal, terminal A become positive, terminal B become negative and center tap is grounded (zero volts). The positive terminal A is connected to the p-side of the diode D1 and the negative terminal B is connected to the n-side of the diode D1. So the diode D1

is forward biased during the positive half cycle and allows electric current through it.

- On the other hand, the negative terminal B is connected to the p-side of the diode D_2 and the positive terminal A is connected to the n-side of the diode D_2. So the diode D_2 is reverse biased during the positive half cycle and does not allow electric current through it.
- The diode D_1 supplies DC current to the load R_L. The DC current produced at the load R_L will return to the secondary winding through a center tap.
- During the **negative half-cycle** of the input, the anode of D_1 becomes negative and the anode of D_2 becomes positive. Hence, D_1 does not conduct and D_2 conducts. The load current flows through D_2 and the voltage drop across R_L will be equal to the input voltage.

Output Waveforms



Expression for average output voltage and current

The average voltage or d.c. voltage available across the load resistance is

$$V_{\text{d.c.}} = \frac{1}{\pi}\int_0^{\pi} V_m \sin \omega t \, \mathrm{d}(\omega t)$$

$$= \frac{V_m}{\pi}[-\cos \omega t]_0^{\pi} = \frac{2V_m}{\pi}$$

$$I_{\text{d.c.}} = \frac{V_{\text{d.c.}}}{R_L} = \frac{2V_m}{\pi R_L} = \frac{2I_m}{\pi} \text{ and } I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$

Full-wave rectification efficiency is

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{(2I_m/\pi)^2 R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 (r_f + R_L)}$$

$$= \frac{8}{\pi^2} \times \frac{R_L}{(r_f + R_L)} = \frac{0.812\ R_L}{r_f + R_L} = \frac{0.812}{1 + \frac{r_f}{R_L}}$$

The efficiency will be maximum if r_f is negligible as compared to R_L.

∴ Maximum efficiency = 81.2%

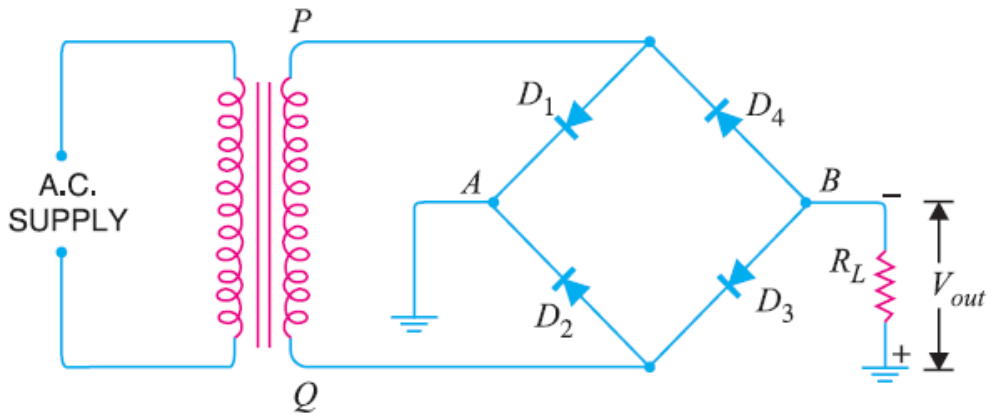
This is double the efficiency due to half-wave rectifier. Therefore, a full-wave rectifier is twice as effective as a half-wave rectifier.

3. Explain the operation of **bridge rectifier** & draw the **wave forms & derive the expression for output voltage & current**.

Full-wave rectifier without transformer (Bridge rectifier).

A bridge rectifier is a type of full wave rectifier which uses four or more diodes in a bridge circuit configuration to efficiently convert the Alternating Current (AC) into Direct Current (DC).

The need for a centre tapped power transformer is eliminated in the bridge rectifier. It contains four diodes *D*1, *D*2, *D*3 and *D*4 connected to form bridge as shown in Fig. 6.26. The a.c. supply to be rectified is applied to the diagonally opposite ends of the bridge through the transformer. Between other two ends of the bridge, the load resistance R_L is connected.

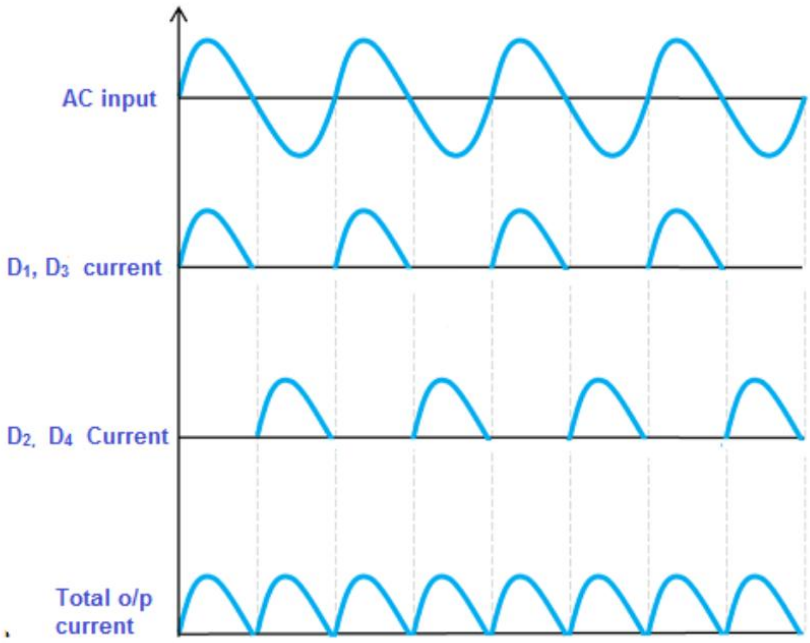


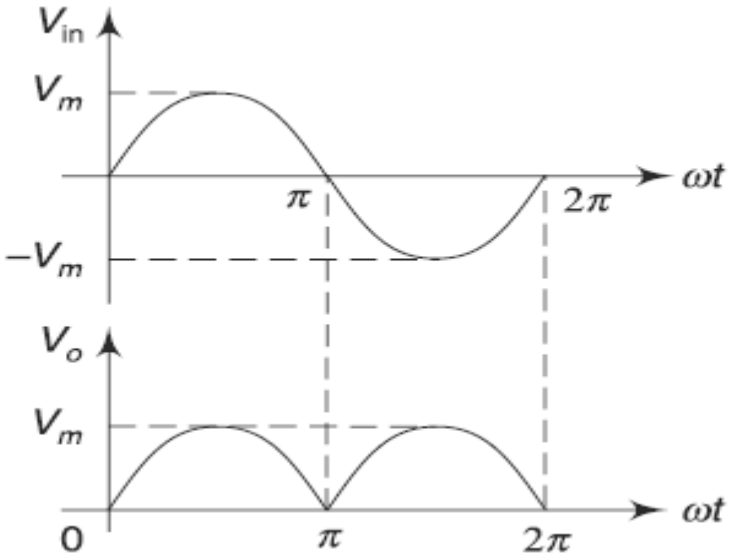
Working Operation of Bridge rectifier

- When input AC signal is applied across the bridge rectifier, during the positive half cycle diodes D_1 and D_3 are forward biased and allows electric current while the diodes D_2 and D_4 are reverse biased and blocks electric current. On the other hand, during the negative half cycle diodes D_2 and D_4 are forward biased and allows electric current while diodes D_1 and D_3 are reverse biased and blocks electric current.

- During the positive half cycle, the terminal A becomes positive while the terminal B becomes negative. This causes the diodes D_1 and D_3 forward biased and at the same time, it causes the diodes D_2 and D_4 reverse biased.

Output Waveforms





- The average values of output voltage and load current for bridge rectifier are the same as for a center tapp full-wave rectifier. Hence,

$$V_{d.c.} = \frac{2V_m}{\pi} \quad \text{and} \quad I_{d.c.} = \frac{V_{d.c.}}{R_L} = \frac{2V_m}{\pi R_L} = \frac{2I_m}{\pi}$$

- The maximum efficiency of a bridge rectifier is 81.2% and the ripple factor is 0.48. The PIV is V_m.

4. What is **Adder**? explain **half adder & full adder** and draw the corresponding **truth table & logic diagram**.

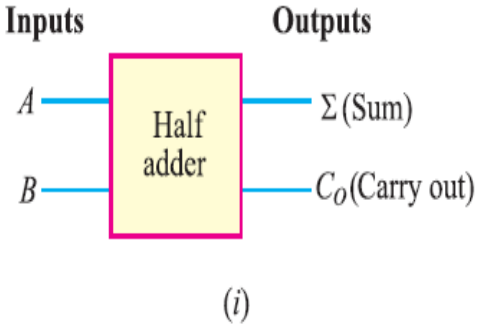
A logic circuit that performs the function of binary addition is called **electronic adder** *or* **adder.** The adder circuit consists of properly connected logic gates. There are usually two forms of the adder in common applications *viz.*

(i) half adder
(ii) full adder.

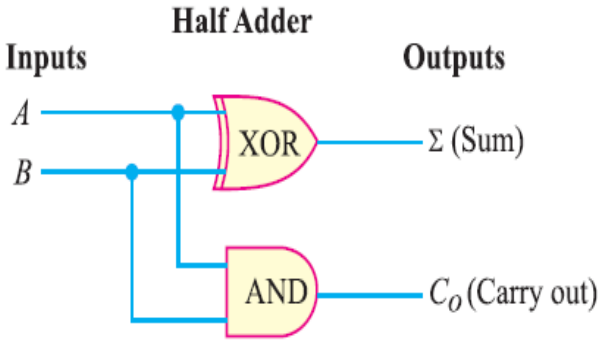
Half Adder (HA).

A logic circuit that can add two binary bits is called a half adder (*HA*).

Figure shows the block symbol for the half adder. The adder circuit would need two inputs and two outputs. The two inputs are for the two digits to be added, either 0 or 1. One output terminal is for the sum of the two inputs and the other output is for the carry, if necessary.



(*i*)



The truth table has two output columns, one column for the sum and one column for the carry. The sum column is labeled with summation symbol Σ. The carry column is labeled with *Co*. The *Co* stands for carry output or *carry out*. Thus the half adder circuit has two inputs (*A*, *B*) and two outputs (Σ, *Co*).

Truth Table

Inputs		Outputs	
B	*A*	Σ	C_O
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1
Binary digits to be added		Sum	Carry out
		XOR	AND

Output columns (sum and carry) can be produced by using two gates as under :

(*a*) The sum column is the output of XOR gate. Remember that XOR gate has HIGH output when either input is HIGH but not when both inputs are the same.

(*b*) The carry column is the output of the AND gate. Both inputs must be HIGH for there to be a HIGH in the output.

Figures 8.15(a) and 8.15(b) shows the block diagram and a truth table of a half-adder. From the truth table, it is noted that the sum output is equivalent to the output of the two input 'XOR' gate. Hence, the sum output can be written as follows:

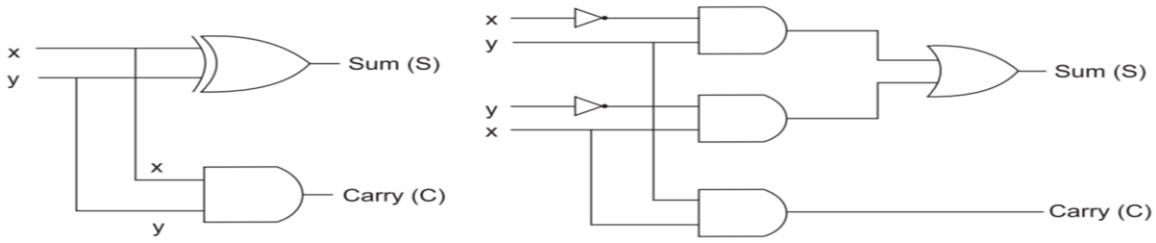
$$Sum(S) = x \oplus y$$

$$Sum(S) = x\,\bar{y} + y\,\bar{x} \qquad ...(8.5)$$

Similarly, the carry output is equivalent to the output of the two input 'AND' gate. Hence, the carry output can be written as

$$Carry\ (C) = xy \qquad ...(8.6)$$

A half-adder can be implemented in terms of logic gates by using the equation (8.5) and (8.6).

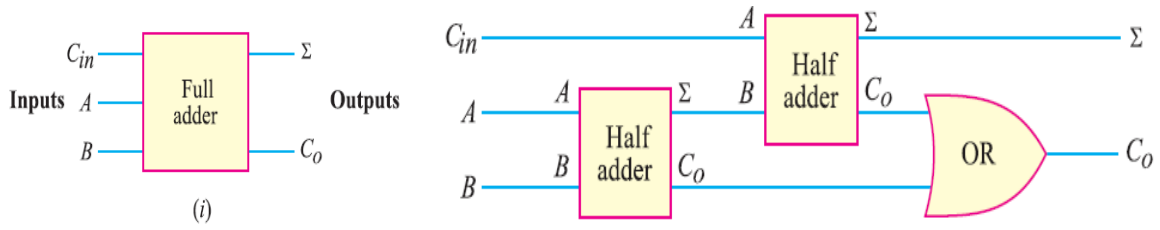


(a) Half adder using XOR gate **(b) Half adder using AOI logic**

Figure (8.16): *Logic diagram of half-adder*

Full Adder (FA).

A full adder adds two binary bits plus a carry input (C_{in}) to produce the sum (Σ) and carry (C_o) outputs. Fig.(*i*) shows the block diagram of a full adder. It is formed by using two half adder circuits and an OR gate as shown in Fig. (*iii*). Note the carry-in input which requires the extra half adder. The output of the OR gate forms the carry-out (*Co*) output.



The full adder has three inputs : C_{in}, *A* and *B*. These three inputs must be added to get the Σ and *Co* outputs. Fig. (*iii*) shows the truth table for a full adder. It shows all the possible combinations of *A, B* and C_{in}.

Truth Table

Inputs			Outputs	
C_{in}	*B*	*A*	Σ	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1
Carry + *B* + *A*			Sum	Carry out

A full adder can be implemented in 2 ways with the help of the equations (8.7) and (8.8) as shown in figures (8.19) and (8.20).

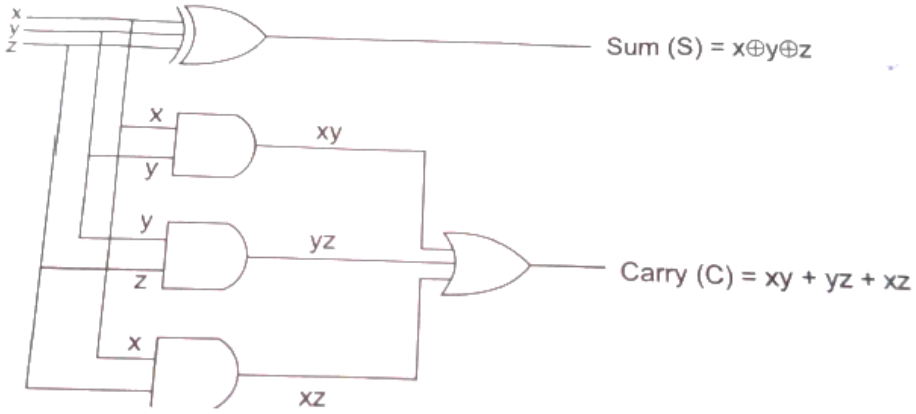


Figure (8.19): *Logic diagram of a full adder using AOI logic*

Sum(S)

$$= \bar{x}\,\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\,\bar{z} + xyz$$
$$= \bar{x}\,\bar{y}z + xyz + \bar{x}y\bar{z} + x\bar{y}\,\bar{z}$$
$$= \bar{z}(x \oplus y) + z(x \Theta y)$$
$$= \bar{z}(x \oplus y) + z\overline{(x \oplus y)} \quad [\because (x \Theta y) = \overline{(x \oplus y)}]$$

Let $A = (x \oplus y)$, $B = z$

$$= A\bar{B} + B\bar{A}$$
$$= A \oplus B$$
$$= (x \oplus y) \oplus z$$

$\therefore$ *Sum*$(S) = (x \oplus y \oplus z)$ ...(8.7)

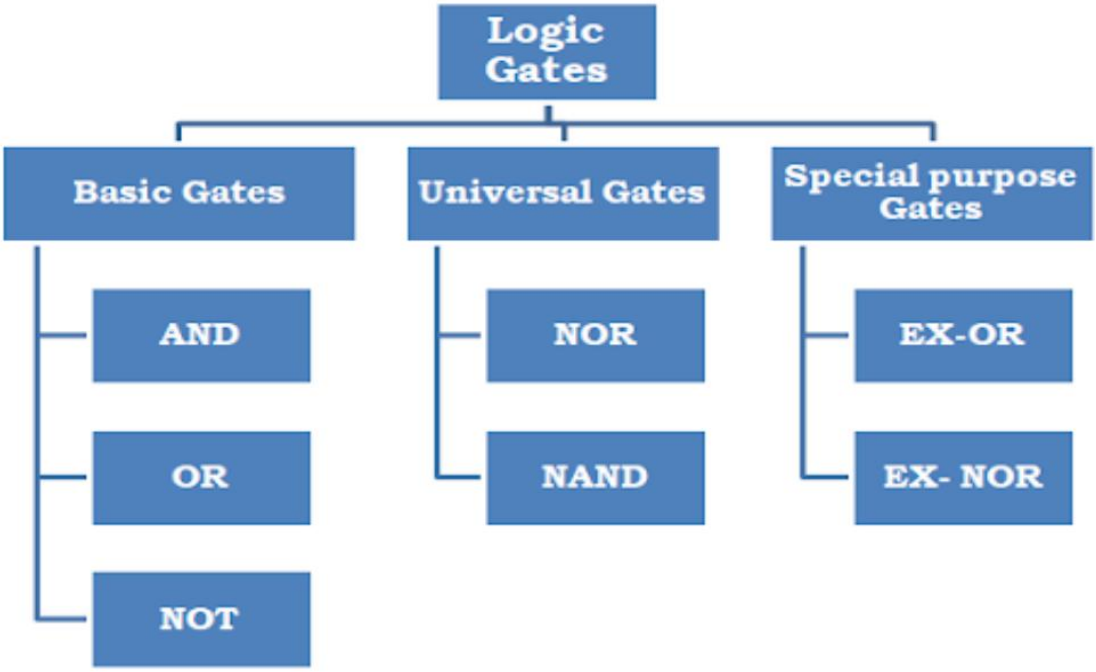
Similarly, *Carry*$(C) = \bar{x}yz + x\bar{y}z + xy\bar{z} + xyz$

$$= \bar{x}yz + x\bar{y}z + xy(z + \bar{z})$$
$$= \bar{x}yz + x\bar{y}z + xy \quad [\because z + \bar{z} = 1]$$
$$= xy + z(\bar{x}y + x\bar{y})$$
$$= xy + z(x \oplus y)$$
$$= (xy + z)(xy + (x \oplus y)) \quad [\because A + BC = (A + B)(A + C]$$
$$= (xy + z)(xy + x\bar{y} + y\bar{x})$$
$$= (xy + z)[x(y + \bar{y}) + y\bar{x}]$$
$$= (xy + z)(x + y\bar{x}) \quad [\because y + \bar{y} = 1]$$
$$= (xy + z)(x + y)(x + \bar{x}) \quad [\because A + BC = (A + B)(A + C)]$$
$$= (x + y)(xy + z) \quad [\because x + \bar{x} = 1]$$
$$= x \cdot x \cdot y + x \cdot y \cdot y + xz + yz$$
$$= xy + xy + xz + yz \quad [\therefore x \cdot x = x,\ y \cdot y = y]$$
$$= xy + yz + zx \quad [\therefore xy + xy = xy]$$

Carry$(C) = xy + yz + zx$ ...(8.8)

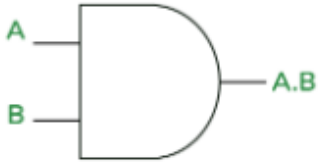
5. Explain different types of logic gates with truth table.

A digital circuit with one or more input signals but only one output signal is called a logic gate.



1. AND Gate

- An AND gate may have two or more inputs but only one output. For the AND gate, there will be four possible combinations ($2^2 = 4$) of inputs such as '00', '01', '10', and '11'.
- When at all inputs are high (1) the output will be high (1).

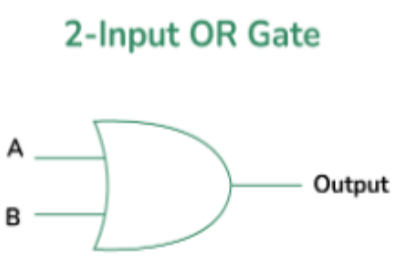


Truth Table

A (Input 1)	B (Input 2)	X = (A.B)
0	0	0
0	1	0
1	0	0
1	1	1

2. OR gate

- The symbol for the 'OR' operation is '+'.
- If the input variables are 'A' and 'B' then the output expression can be represented by Y = A + B.
- When one or more of the inputs is high (1) the output will be high (1).



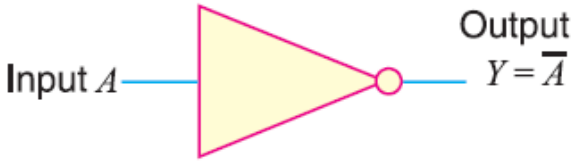
Truth Table

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

3. NOT Gate

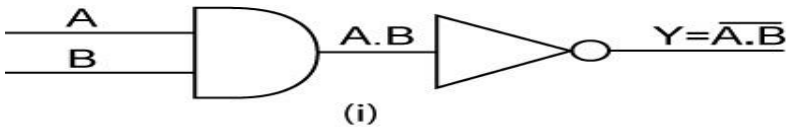
- 'NOT' gate is also called an inverter.
- It has only one input and one output.
- The output is the opposite to the input.

A	Y
0	1
1	0

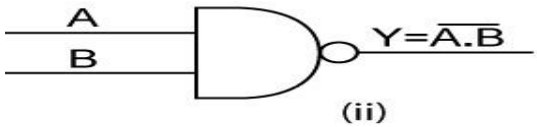


4. NAND Gate

- The 'NAND' Gate is constructed by using an 'AND' gate and a 'NOT' gate.
- "NOT AND", hence when at least one input is high (1) the output is high(1). If both inputs are high (1) the the output is low (0).



(i)

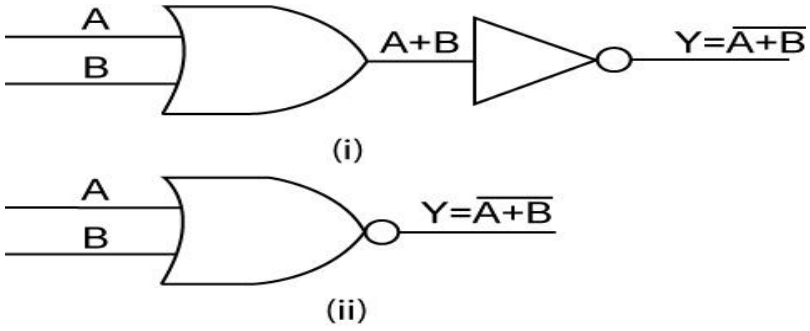


(ii)

Input		output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

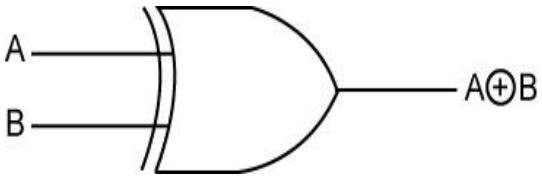
5. NOR Gate

- The NOR gate is constructed by using a 'OR' gate and a NOT gate.
- The NOR gate is obtained by placing a NOT gate at the output of a 'OR' gate.



Input		output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

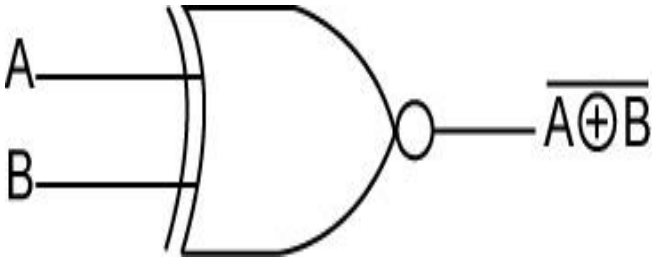
6. **Exclusive-OR(XOR) Gate**
   - From the truth table of two inputs XOR gate, it can say that when the two **input variables are different** such as '01' and '10', then the output 'Y' will be '1'.
   - When the two **input variables are equal** such as '00' and '11', then the output 'Y' will be '0'.
   - The symbol '⊕' denotes 'Exclusive-OR' operation



Input		output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

7. **Exclusive-NOR(XNOR) Gate**

- An exclusive-NOR gate or XNOR gate can be obtained by using XOR gate and a NOT gate.
- From the truth table of two inputs XNOR gate, it may be noted that , when the two input variables are different such as '01' and '10', the output "Y" will be "0"
- When the two input variables are equal such as '00' and '11', then the output 'Y' will be '1'.



Input		output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

6. Problem on hamming code.

- See the Unit 5 PPT with both encode and decode of hamming code

7. Problem on general number systems.

- See the Unit 5 PPT with all the number system conversions

8. Problem on Excess-3 & Gray code.

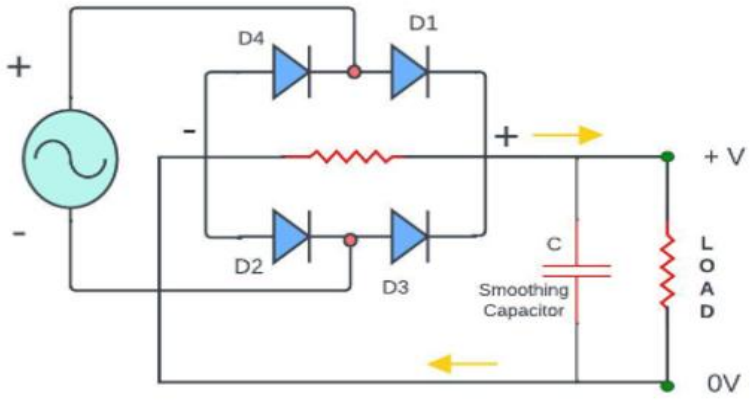
- See the Unit 5 PPT for both XS-3 code and Gray code conversions with steps.

9. Explain L-filter & C-filter and draw the corresponding wave forms.

The output of a rectifier contains d.c. component as well as a.c. component. Filters are used to minimise the undesirable a.c., i.e. ripple leaving only the d.c. component to appear at the output.

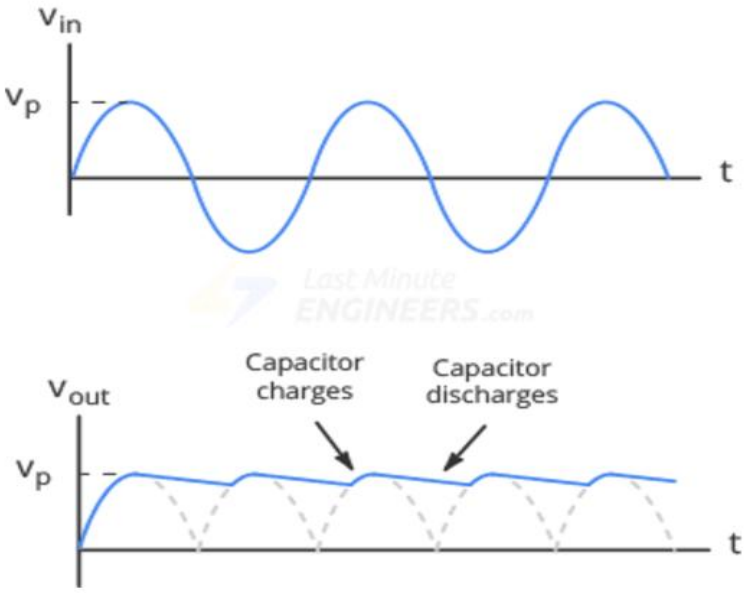
(i) **Capacitor filter.**

It consists of a capacitor C placed across the rectifier output in parallel with load R_L. The pulsating direct voltage of the rectifier is applied across the capacitor.



Working Operation of Capacitor filter

- During the positive half-cycle of the AC input one pair of diodes (D1 & D2) are conducts that allowing the capacitor to charge to peak value of the rectified voltage.
- During the negative half-cycle the other one diode pair (D3 & D4) are conducts that allowing the capacitor to discharge into the load.



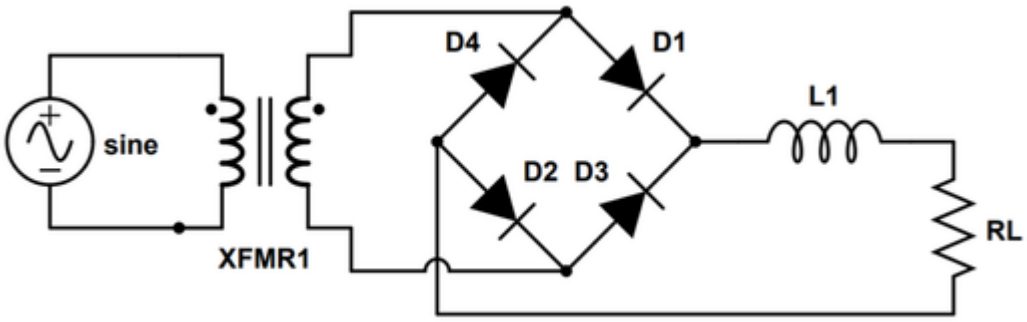
full Wave Bridge Rectifier with Capacitor Filter waveform

Output voltage:

- The **voltage across the capacitor** helps to smooth out of the pulsating DC obtained from the full-wave rectifier.
- The output voltage is the sum of the DC components and reduced ripple voltage due to the capacitor filter.

ii) Inductor Filter or Choke filter

In this type of filter, an inductor (choke) is connected in series with the load. It is known that the **inductor opposes change in the current**. So, the ripple which is change in the current is opposed by the inductor and it tries to smoothen the output.



Working Operation of L-filter

- Positive Half-Cycle of Input: Diodes D1 and D2 conduct, providing a path for current flow through the inductor L1 and the load RL.
- Negative Half-Cycle of Input: Diodes D3 and D4 conduct, providing a path for current flow through the inductor L1 and the load RL.

Filtering Action of L1:

- The pulsating DC voltage from the bridge rectifier has a DC component and an AC component (the ripple).
- The inductor L1 offers a low opposition (ideally zero resistance) to the DC component of the current, allowing it to pass through to the load RL with minimal loss.
- The inductor L1 offers a high opposition (high reactance) to the AC component (ripple) because the ripple frequency is high.

